Reinforcement Learning

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Recap

- Sometimes, it is better to learn a stochastic policy than a deterministic policy
- Policy gradient methods suffer from poor sample efficiency
- Importance sampling can help alleviate this problem
- However, we need to make sure that current policy is not far from policy with which we have collected trajectories
- Small changes in the policy parameters can unexpectedly lead to big changes in the policy.
- TRPO uses importance sampling to take multiple gradient steps and constrains the optimization objective in the policy space
- PPO does the same but approximately enforces KL constraint without computing natural gradients.

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- So far, we have only looked at model-free reinforcement learning

What is a Model?

- A model $\mathcal{M}$ is a representation of an MDP $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R} \rangle$
- We will assume state space $\mathcal{S}$ and action space $\mathcal{A}$ are known
- So a model $\mathcal{M} = \langle \mathcal{P}_\phi, \mathcal{R}_\phi \rangle$ represents state transitions $\mathcal{P}_\phi \approx \mathcal{P}$ and rewards $\mathcal{R}_\phi \approx \mathcal{R}$

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- We can also only learn one of state transitions and reward model

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- We pick a loss function like MSE, KL Divergence etc., and then optimize

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- Advantages:
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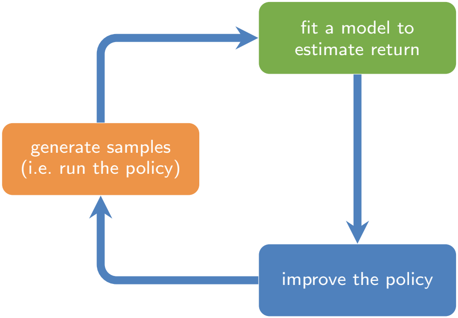
  - Can reason about model uncertainty
- Disadvantages:
  - First learn a model, then construct a value function. This means we will have two sources of approximation error

Model-Based RL



estimate $p(\mathbf{s}' \mid \mathbf{s}, \mathbf{a})$ (model-based)

supervised learning

$$\min_{\phi} \sum_{i} \| f_{\phi}(\mathbf{s}_i, \mathbf{a}_i) - \mathbf{s}'_i \|^2$$

optimize $\pi_\theta(\mathbf{a} \mid \mathbf{s})$ (model-based)

- So, now we have the following algorithm
  1. Run some policy (e.g. random policy) to collect data $D = \{(s, a, s')_i\}$
  2. Learn model $f_\phi(s, a)$ to minimize $\sum_i \| f_\phi(s_i, a_i) - s_i' \|$
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- Thus, it will learn an incomplete policy and fall at the top.
- Data distribution mismatch $p_{\pi_0}(s) \neq p_{\pi_f}(s)$

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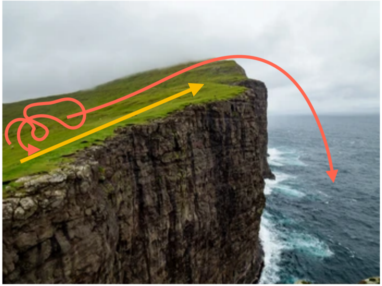
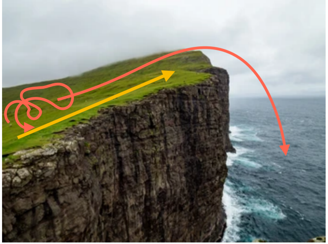
- Now, we update the model using data collected with planning
- Then, replan periodically to help account for mistakes.

Revisiting the Cliff





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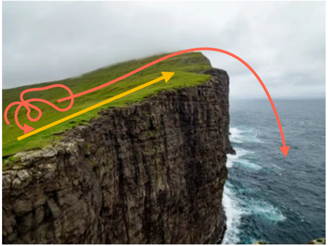

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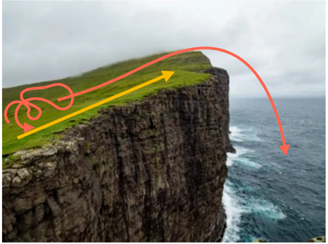

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- Then we re-train the model using the new observations
- The final policy is different now. It learns to go to the top and stop.

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- Algorithm:
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  5. Go to step 1
- Compatible with variety of model-free RL methods
- Could additionally use D_{env} in policy update (step 4)

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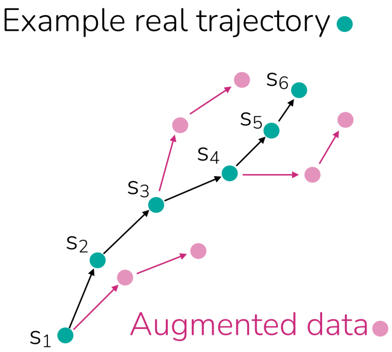

When to use model-based RL?

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  - Models are immensely useful if easy to learn
  - Model can be trained without reward labels (self-supervised)
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Whether to use a model depends on how hard it is to learn!

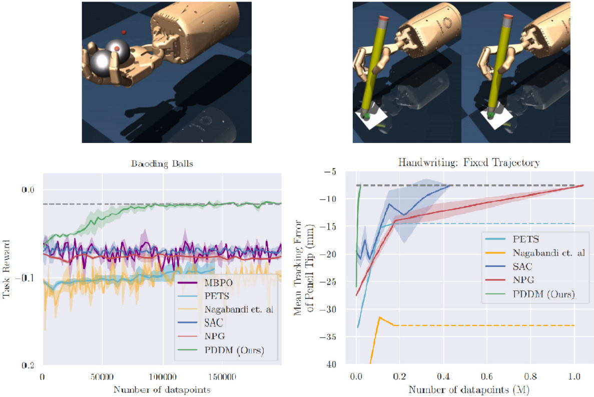
Case study: Model-based RL for dexterous manipulation

Model-free methods:

- **SAC**: actor-critic method
- **NPG**: policy gradient method

Model-based methods:

- **PDDM**: proposed method
- **MBPO**: RL with model-generated data
- **PETS**: CEM-based planner
- **Nagabandi et al.**: random shooting, no ensembles



Nagabandi, et al. Deep Dynamics Models for Learning Dexterous Manipulation

Summary

- Reinforcement Learning can be categorised into Model-Based and Model-Free.
- In Model-Based RL, we also learn the model of the world i.e, the state transition model and/or the reward model
- Model-Based RL can be employed with both value and/or policy based methods
- We can use the model to augment the real environment data
- Models can be immensely useful if easy to learn but sometimes they can be harder to learn than a policy
- Whether to use a model depends on how hard it is to learn

References

These slides have been adapted from

- Chelsea Finn & Karol Hausman, Stanford CS224R: Deep Reinforcement Learning
- Sergey Levine, Berkeley CS285 Deep Reinforcement Learning
- David Silver, Deepmind: Introduction to Reinforcement Learning